

1. Write a Linux command to display the operating system name, kernel version and machine architecture.

Command: _____

What information was displayed?

2. Execute the following command:

`uname -a`

Identify the following from the output.

Kernel name: _____

Kernel version: _____

Machine architecture: _____

3. Which command can be used to display the amount of used and available memory in a human-readable format?

Command: _____

What does the option indicate?

4. Execute `free -h`. Record the values reported by your system.

Total memory: _____

Used memory: _____

Available memory: _____

5. Write the command used to display disk-space usage for mounted file systems in a human-readable format.

Command: _____

6. Execute `df -h`.

Identify any one mounted file system from the output.

Filesystem: _____

Size: _____

Used: _____

Available: _____

Mounted on: _____

7. Which command can be used to display the processes currently running in your terminal session?

Command: _____

8. Run `ps` and observe its output.

What is PID?

What PID is assigned to your shell?

9. Execute top.

Write any four types of information displayed by the command.

- a. _____
- b. _____
- c. _____
- d. _____

10. Compare ps and top. Complete the table.

Feature		ps		top
Output type		_____		_____
Continuously updated?		_____		_____
Shows CPU usage?		_____		_____
Useful for live monitoring?		_____		_____

11. Write the command used to display users who are currently logged into the system.

Command: _____

Execute it and record the number of logged-in users.

Number of users: _____

12. The ls command is executed in a directory.

What is its purpose?

Write a command to display files with detailed information.

Command: _____

13. You need to identify the modules currently loaded into the Linux kernel.

Which command should be used?

Command: _____

What is a kernel module?

14. Consider the command:

modinfo nvidia

What type of information would be displayed?

How is modinfo different from lsmod?

15. A program needs to be examined to determine which system calls are being made.

Complete the command:

_____ echo "Hello BCA"

What is the purpose of this command?

16. Execute:

strace -c ls

What is the purpose of the -c option?

Record the three system calls with the highest number of calls.

1. _____

2. _____

3. _____

4. Study the following command:

strace -f bash -c 'ls | grep py'

What does -f instruct strace to do?

Why is this option useful for this command?

17. Command Challenge

A system administrator wants to perform the following tasks. Write the most appropriate command for each task.

a. Check the Linux kernel version:

b. Check available RAM:

c. Check available disk space:

d. View currently running processes:

e. Monitor processes continuously:

f. Check logged-in users:

g. View loaded kernel modules:

h. Trace the system calls generated by ls:

i. Generate an SSH key pair:

j. Connect to user student on server 10.0.0.25:
